

STANDARD FORMS OF BOOLEAN EXPRESSIONS

All Boolean expressions, regardless of their form, can be converted into either of two standard forms: the sum-of-products form or the product-of sums form. Standardization makes the evaluation, simplification, and implementation of Boolean expressions much more systematic and easier.

THE SUM-OF-PRODUCTS (SOP)

The Sum-of-Products (SOP) Form When two or more product terms are summed by Boolean addition, the resulting expression is a sum-of-products (SOP).

Some examples are:

$$AB + ABC$$

$$ABC + CDE + BCD$$

$$AB + BCD + AC$$

Also, an SOP expression can contain a single-variable term, as in $A + ABC + BCD$.

THE STANDARD SOP FORM

So far, you have seen SOP expressions in which some of the product terms do not contain all of the variables in the domain of the expression.

For example, the expression $ABC + ABD + ABCD$ has a domain made up of the variables A, B, C, and D. However, notice that the complete set of variables in the domain is not represented in the first two terms of the expression; that is, D or D is missing from the first term and C or C is missing from the second term.

A standard SOP expression is one in which all the variables in the domain appear in each product term in the expression. For example, $ABCD + ABCD + ABCD$ is a standard SOP expression.

THE PRODUCT-OF-SUMS (POS)

A sum term was defined before as a term consisting of the sum (Boolean addition) of literals (variables or their complements). When two or more sum terms are multiplied, the resulting expression is a product-of-sums (POS).

Some examples are $(A + B)(A + B + C)$ $(A + B + C)(C + D + E)(B + C + D)$ $(A + B)(A + B + C)(A + C)$

THE STANDARD POS FORM

So far, you have seen POS expressions in which some of the sum terms do not contain all of the variables in the domain of the expression.

For example, the expression $(A + B + C)(A + B + D)(A + B + C + D)$ has a domain made up of the variables A, B, C, and D.

A standard POS expression is one in which all the variables in the domain appear in each sum term in the expression. For example, $(A + B + C + D)(A + B + C + D)(A + B + C + D)$

CONVERTING PRODUCT TERMS TO STANDARD SOP

Each product term in an SOP expression that does not contain all the variables in the domain can be expanded to standard SOP to include all variables in the domain and their complements. As stated in the following steps, a nonstandard SOP expression is converted into standard form using Boolean algebra rule 6 ($A + \bar{A} = 1$): A variable added to its complement equals

1. Step 1. Multiply each nonstandard product term by a term made up of the sum of a missing variable and its complement. This results in two product terms. As you know, you can multiply anything by 1 without changing its value.

Step 2. Repeat Step 1 until all resulting product terms contain all variables in the domain in either complemented or complemented form. In converting a product term to standard form, the number of product terms is doubled for each missing variable.

EXAMPLE

Convert the following Boolean expression into standard SOP form: $ABC + AB + ABCD$

Solution The domain of this SOP expression A, B, C, D. Take one term at a time.

The first term, ABC , is missing variable D or $\bar{D}$, so multiply the first term by $(D + \bar{D})$ as follows: $ABC = ABC(D + \bar{D}) = ABCD + ABC\bar{D}$

In this case, two standard product terms are the result. The second term, AB , is missing variables C or $\bar{C}$ and D or $\bar{D}$, so first multiply the second term by $C + \bar{C}$ as follows: $AB = AB(C + \bar{C}) = ABC + AB\bar{C}$

The two resulting terms are missing variable D or $\bar{D}$, so multiply both terms by $D + \bar{D}$ as follows:

$$\begin{aligned}\bar{A}\bar{B} &= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} = \bar{A}\bar{B}C(D + \bar{D}) + \bar{A}\bar{B}\bar{C}(D + \bar{D}) \\ &= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}\end{aligned}$$

In this case, four standard product terms are the result.

The third term, $AB\bar{C}D$, is already in standard form. The complete standard SOP form of the original expression is as follows:

$$\bar{A}\bar{B}C + \bar{A}\bar{B} + AB\bar{C}D = \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + AB\bar{C}D$$

Converting Sum Terms to Standard POS

Converting a Sum Term to Standard POS

Each sum term in a POS expression that does not contain all the variables in the domain can be expanded to standard form to include all variables in the domain and their complements. As stated in the following steps, a nonstandard POS expression is converted into standard form using Boolean algebra rule 8 ($A \cdot \bar{A} = 0$) from Table 4–1: A variable multiplied by its complement equals 0.

- Step 1:** Add to each nonstandard product term a term made up of the product of the missing variable and its complement. This results in two sum terms. As you know, you can add 0 to anything without changing its value.
- Step 2:** Apply rule 12 from Table 4–1: $A + BC = (A + B)(A + C)$
- Step 3:** Repeat Step 1 until all resulting sum terms contain all variables in the domain in either complemented or uncomplemented form.

Convert the following Boolean expression into standard POS form:

$$(A + \bar{B} + C)(\bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

Solution

The domain of this POS expression is A, B, C, D . Take one term at a time. The first term, $A + \bar{B} + C$, is missing variable D or $\bar{D}$, so add $D\bar{D}$ and apply rule 12 as follows:

$$A + \bar{B} + C = A + \bar{B} + C + D\bar{D} = (A + \bar{B} + C + D)(A + \bar{B} + C + \bar{D})$$

The second term, $\bar{B} + C + \bar{D}$, is missing variable A or $\bar{A}$, so add $A\bar{A}$ and apply rule 12 as follows:

$$\bar{B} + C + \bar{D} = \bar{B} + C + \bar{D} + A\bar{A} = (A + \bar{B} + C + \bar{D})(\bar{A} + \bar{B} + C + \bar{D})$$

The third term, $A + \bar{B} + \bar{C} + D$, is already in standard form. The standard POS form of the original expression is as follows:

$$(A + \bar{B} + C)(\bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D) = \\ (A + \bar{B} + C + D)(A + \bar{B} + C + \bar{D})(A + \bar{B} + C + \bar{D})(\bar{A} + \bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

Converting SOP Expressions to Truth Table Format

Develop a truth table for the standard SOP expression $\overline{A}\overline{B}C + A\overline{B}\overline{C} + ABC$.

Solution

There are three variables in the domain, so there are eight possible combinations of binary values of the variables as listed in the left three columns of Table 4–6. The binary values that make the product terms in the expressions equal to 1 are

TABLE 4–6				
Inputs			Output	Product Term
A	B	C	X	
0	0	0	0	
0	0	1	1	$\overline{A}\overline{B}C$
0	1	0	0	
0	1	1	0	
1	0	0	1	$A\overline{B}\overline{C}$
1	0	1	0	
1	1	0	0	
1	1	1	1	ABC

Converting POS Expressions to Truth Table Format

Determine the truth table for the following standard POS expression:

$$(A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + B + \overline{C})(\overline{A} + \overline{B} + C)$$

Solution

There are three variables in the domain and the eight possible binary values are listed in the left three columns of Table 4–7. The binary values that make the sum terms in the expression equal to 0 are $A + B + C$: 000; $A + \overline{B} + C$: 010; $A + \overline{B} + \overline{C}$: 011; $\overline{A} + B + \overline{C}$: 101; and $\overline{A} + \overline{B} + C$: 110. For each of these binary values, place a 0 in the output column as shown in the table. For each of the remaining binary combinations, place a 1 in the output column.

TABLE 4–7				
Inputs			Output	Sum Term
A	B	C	X	
0	0	0	0	$(A + B + C)$
0	0	1	1	
0	1	0	0	$(A + \overline{B} + C)$
0	1	1	0	$(A + \overline{B} + \overline{C})$
1	0	0	1	
1	0	1	0	$(\overline{A} + B + \overline{C})$
1	1	0	0	$(\overline{A} + \overline{B} + C)$
1	1	1	1	

From the truth table in Table 4–8, determine the standard SOP expression and the equivalent standard POS expression.

TABLE 4–8			
Inputs			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Solution

There are four 1s in the output column and the corresponding binary values are 011, 100, 110, and 111. Convert these binary values to product terms as follows:

$$011 \longrightarrow \bar{A}BC$$

$$100 \longrightarrow A\bar{B}\bar{C}$$

$$110 \longrightarrow AB\bar{C}$$

$$111 \longrightarrow ABC$$

The resulting standard SOP expression for the output X is

$$X = \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C} + ABC$$

For the POS expression, the output is 0 for binary values 000, 001, 010, and 101. Convert these binary values to sum terms as follows:

$$000 \longrightarrow A + B + C$$

$$001 \longrightarrow A + B + \bar{C}$$

$$010 \longrightarrow A + \bar{B} + C$$

$$101 \longrightarrow \bar{A} + B + \bar{C}$$

The resulting standard POS expression for the output X is

$$X = (A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + \bar{C})$$

Minterms and maxterms are related

- Any minterm m_i is the *complement* of the corresponding maxterm M_i

Minterm	Shorthand	Maxterm	Shorthand
$x'y'z'$	m_0	$x + y + z$	M_0
$x'y'z$	m_1	$x + y + z'$	M_1
$x'yz'$	m_2	$x + y' + z$	M_2
$x'yz$	m_3	$x + y' + z'$	M_3
$xy'z'$	m_4	$x' + y + z$	M_4
$xy'z$	m_5	$x' + y + z'$	M_5
xyz'	m_6	$x' + y' + z$	M_6
xyz	m_7	$x' + y' + z'$	M_7

- For example, $m_4' = M_4$ because $(xy'z')' = x' + y + z$

KARNAUGH MAP

A Karnaugh map is similar to a truth table because it presents all of the possible values of input variables and the resulting output for each value. Instead of being organized into columns and rows like a truth table, the Karnaugh map is an array of cells in which each cell represents a binary value of the input variables. The cells are arranged in a way so that simplification of a given expression is simply a matter of properly grouping the cells. Karnaugh maps can be used for expressions with two, three, four, and five variables

The 3-Variable Karnaugh Map

The 3-variable Karnaugh map is an array of eight cells, as shown in Figure 4–25(a). In this case, A , B , and C are used for the variables although other letters could be used. Binary values of A and B are along the left side (notice the sequence) and the values of C are across the top. The value of a given cell is the binary values of A and B at the left in the same row combined with the value of C at the top in the same column. For example, the cell in the upper left corner has a binary value of 000 and the cell in the lower right corner has a binary value of 101. Figure 4–25(b) shows the standard product terms that are represented by each cell in the Karnaugh map.

$AB \backslash C$	C	
	0	1
00		
01		
11		
10		

(a)

$AB \backslash C$	C	
	0	1
00	$\bar{A}\bar{B}\bar{C}$	$\bar{A}\bar{B}C$
01	$\bar{A}B\bar{C}$	$\bar{A}BC$
11	$AB\bar{C}$	ABC
10	$A\bar{B}\bar{C}$	$A\bar{B}C$

(b)

FIGURE 4–25 A 3-variable Karnaugh map showing Boolean product terms for each cell.

The 4-Variable Karnaugh Map

The 4-variable Karnaugh map is an array of sixteen cells, as shown in Figure 4–26(a). Binary values of A and B are along the left side and the values of C and D are across the top. The value of a given cell is the binary values of A and B at the left in the same row combined with the binary values of C and D at the top in the same column. For example, the cell in the upper right corner has a binary value of 0010 and the cell in the lower right corner has a binary value of 1010. Figure 4–26(b) shows the standard product terms that are represented by each cell in the 4-variable Karnaugh map.

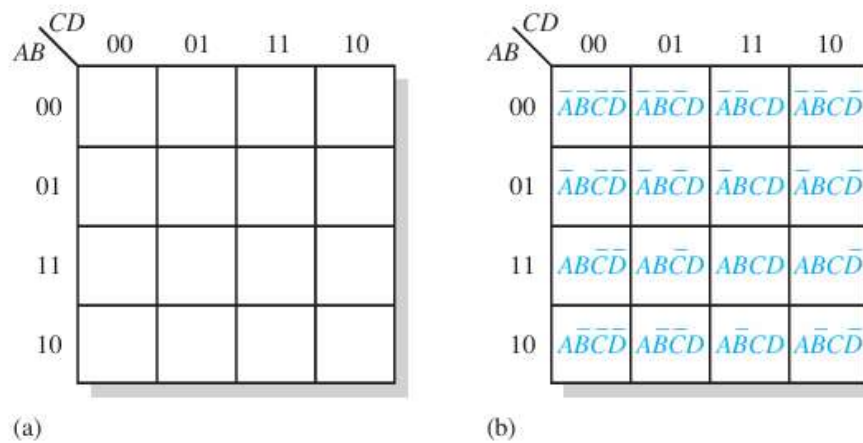


FIGURE 4–26 A 4-variable Karnaugh map.

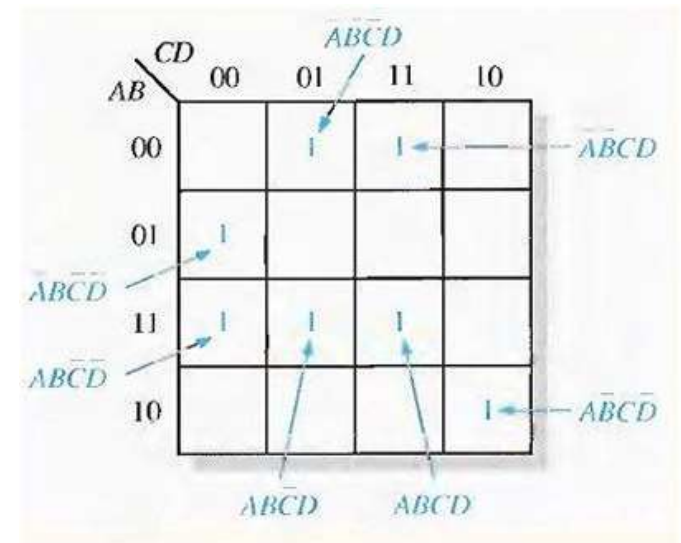
KARNAUGH MAP SOP MINIMIZATION

For an SOP expression in standard form, a 1 is placed on the Karnaugh map for each product term in the expression. Each 1 is placed in a cell corresponding to the value of a product term. For example, for the product term ABC, a 1 goes in the 101 cell on a 3-variable map.

Example

Map the following standard SOP expression on a Karnaugh map:

$$\overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + A\overline{B}C\overline{D} + ABCD + A\overline{B}C\overline{D} + \overline{A}\overline{B}C\overline{D} + A\overline{B}C\overline{D}$$



Karnaugh Map Simplification of SOP Expressions

Grouping the 1s, you can group 1s on the Karnaugh map according to the following rules by enclosing those adjacent cells containing 1s. The goal is to maximize the size of the

groups and to minimize the number of groups.

A group must contain either 1, 2, 4, 8, or 16 cells, which are all powers of two. In the case of a 3-variable map, $2^3 = 8$ cells is the maximum group.

Each cell in a group must be adjacent to one or more cells in that same group.

Always include the largest possible number of 1s in a group in accordance with rule 1.

Each 1 on the map must be included in at least one group. The 1s already in a group can be included in another group as long as the overlapping groups include noncommon 1s.

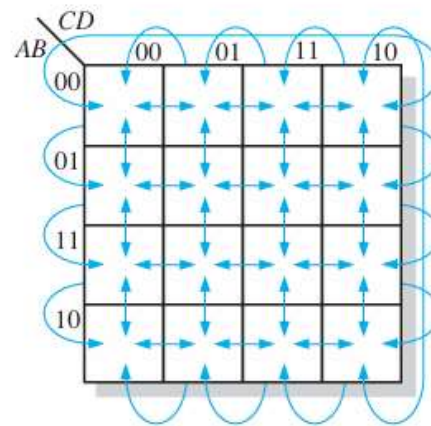


FIGURE 4-27 Adjacent cells on a Karnaugh map are those that differ by only one variable. Arrows point between adjacent cells.

Steps to Solve Expression using K-map

1. Select the K-map according to the number of variables.
2. Identify minterms or maxterms as given in the problem.
3. For SOP put 1's in blocks of K-map respective to the minterms (0's elsewhere).
4. For POS put 0's in blocks of K-map respective to the max terms (1's elsewhere).
5. Make rectangular groups containing total terms in power of two like 2,4,8 ..(except 1) and try to cover as many elements as you can in one group.
6. From the groups made in step 5 find the product terms and sum them up for SOP form.

Advantages of K-MAP

- **Makes Logic Simpler:** It makes complicated Boolean expressions simpler.
- **Minimizes Logic Gates:** Simplifying the logic helps us to use fewer logic gates, making circuits more efficient.
- **Reduce Errors:** The visual representation of k-map helps to avoid errors while simplifying.
- **Time-Saving:** It's quicker than traditional methods for simplifying logic.
- **No need for Boolean Laws:** K-map doesn't require deep knowledge of Boolean laws, making it easy for beginners.

Disadvantages of K-MAP

- **Limited to Fewer Variables:** K-maps are best suited for 2 to 4 variables and above it, process becomes hard and complicated to manage.
- **Not suitable for all functions:** In some cases, its hard to group terms correctly, leading to errors and making simplification difficult.
- **Space Limitations:** As the number of variables increases, the K-map grid becomes too large to handle easily.
- **Requires Careful Grouping:** Sometimes incorrect grouping of terms can cause mistakes in logic simplification.

Example:

Group the 1s in each of the Karnaugh maps in Fig.(5-6).

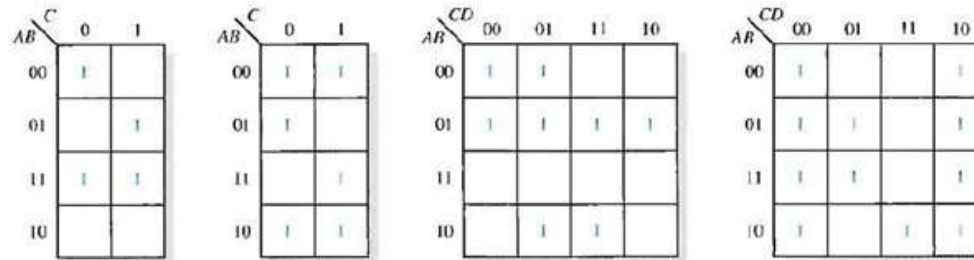


Fig.(5-6)

Solution:

The groupings are shown in Fig.(5-7). In some cases, there may be more than one way to group the 1s to form maximum groupings.

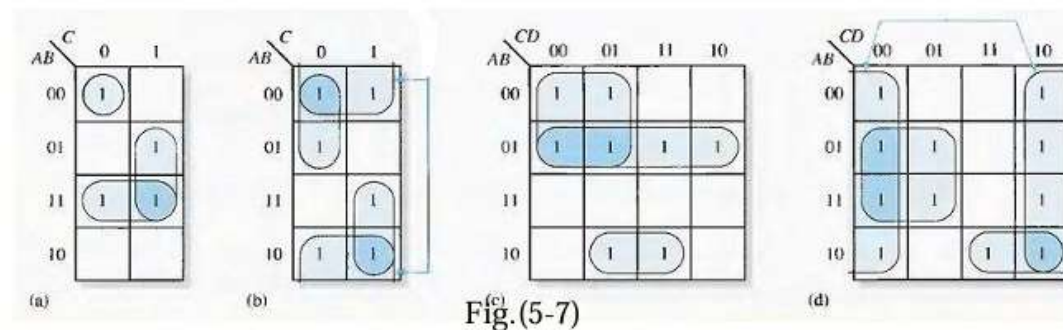
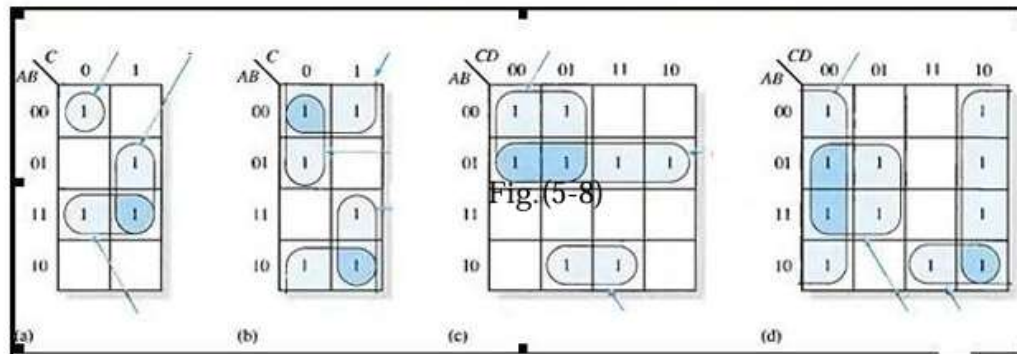


Fig.(5-7)

Example:

Determine the product terms for each of the Karnaugh maps in Fig.(5-7) and write the resulting minimum SOP expression.



Solution:

The resulting minimum product term for each group is shown in Fig.(5-8).

The minimum SOP expressions for each of the Karnaugh maps in the figure are:

$$(a) AB + BC + \overline{A}BC$$

$$(b) \overline{B} + AC + \overline{A}\overline{C}$$

$$(c) \overline{A}B + \overline{A}C + ABD$$

$$(d) \overline{D} + \overline{A}BC + B\overline{C}$$

Example: Use a Karnaugh map to minimize the following standard SOP expression:

$$ABC + ABC + ABC + ABC + ABC$$

Example: Use a Karnaugh map to minimize the following SOP expression:

$$\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}CD + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} + \overline{A}BCD$$

KARNAUGH MAP POS MINIMIZATION

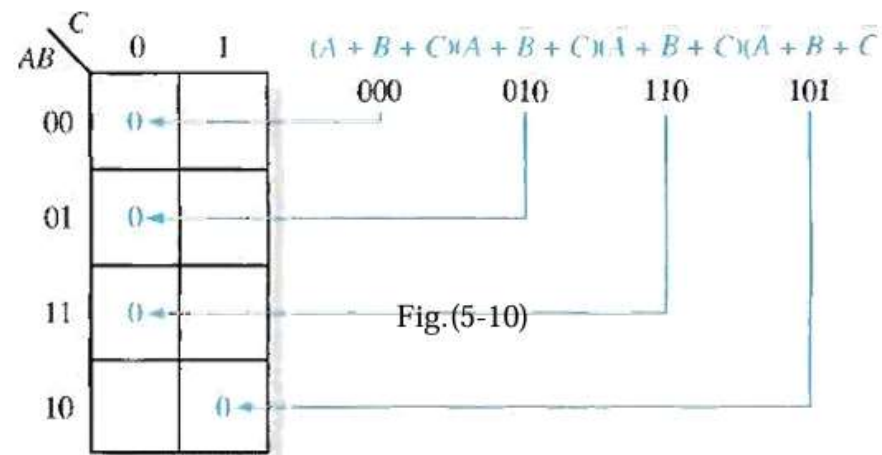
In this section, we will focus on POS expressions. The approaches are much the same except that with POS expressions, 0s representing the standard sum terms are placed on the Karnaugh map instead of 1s.

For a POS expression in standard form, a 0 is placed on the Karnaugh map for each sum term in the expression. Each 0 is placed in a cell corresponding to the value of a sum term. For example, for the sum term $A + B + C$, a 0 goes in the 0 1 0 cell

When a POS expression is completely mapped, there will be a number of 0s on the Karnaugh map equal to the number of sum terms in the standard POS expression. The cells that do not have a 0 are the cells for which the expression is 1. Usually, when working with POS expressions, the 1s are left off. The following steps and the illustration in Fig.(5-10) show the mapping process.

Step 1. Determine the binary value of each sum term in the standard POS expression. This is the binary value that makes the term equal to 0.

Step 2. As each sum term is evaluated, place a 0 on the Karnaugh map in the corresponding cell.



Example of mapping a standard POS expression.

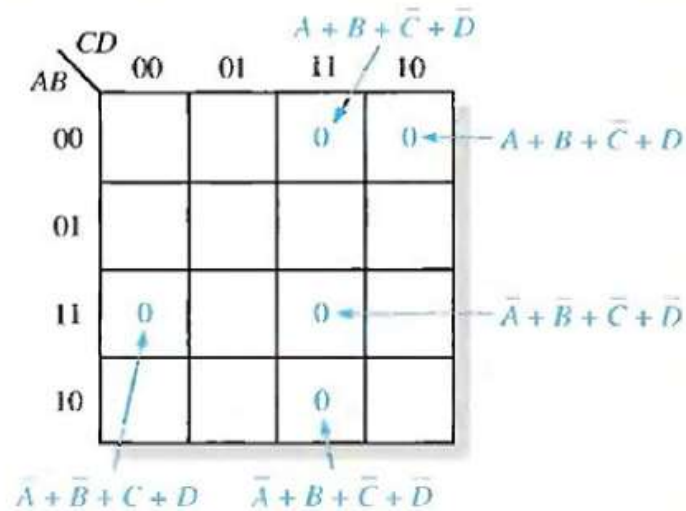
Example:

Map the following standard POS expression on a Karnaugh map:

$$(\bar{A} + \bar{B} + C + D)(\bar{A} + B + \bar{C} + \bar{D})(A + B + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})(A + B + \bar{C} + \bar{D})$$

Solution:

$$\begin{array}{ccccc} (\bar{A} + \bar{B} + C + D) & (\bar{A} + B + \bar{C} + \bar{D}) & (A + B + \bar{C} + D) & (\bar{A} + \bar{B} + \bar{C} + \bar{D}) & (A + B + \bar{C} + \bar{D}) \\ 1100 & 1011 & 0010 & 1111 & 0011 \end{array}$$



Karnaugh Map Simplification of POS Expressions

The process for minimizing a POS expression is basically the same as for an SOP expression except that you group 0s to produce minimum sum terms instead of grouping 1s to produce minimum product terms. The rules for grouping the 0s are the same as those for grouping the 1s that you learned before.

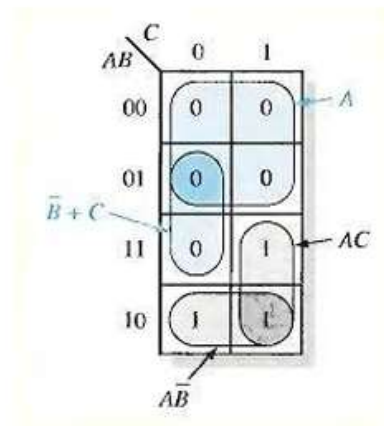
Example:

Use a Karnaugh map to minimize the following standard POS expression:

Also, derive the equivalent SOP expression.

$$(A + B + C)(A + B + \bar{C})(A + \bar{B} + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$$

Solution:



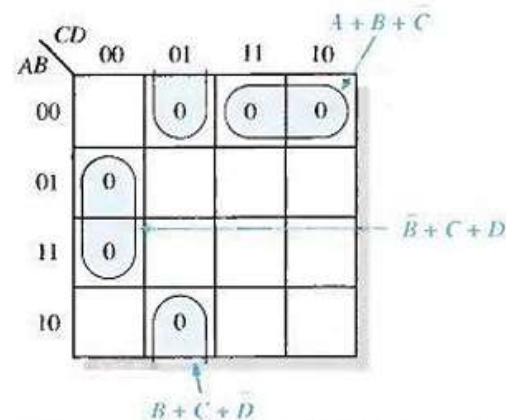
Example: Use a Karnaugh map to minimize the following POS expression:

$$(B + C + D)(A + B + \bar{C} + D)(\bar{A} + B + C + \bar{D})(A + \bar{B} + C + D)(\bar{A} + \bar{B} + C + D)$$

Example: Using a Karnaugh map, convert the following standard POS expression into a minimum POS expression, a standard SOP expression, and

$$(\bar{A} + \bar{B} + C + D)(A + \bar{B} + C + D)(A + B + C + \bar{D})$$

$$(A + B + \bar{C} + \bar{D})(\bar{A} + B + C + \bar{D})(A + B + \bar{C} + D)$$



(a) Minimum POS: $(A + B + C)(\bar{B} + \bar{C} + D)(B + C + \bar{D})$

		$\overline{A}BCD$		$A\overline{B}CD$		
		CD				
AB		00	01	11	10	
	00	1	0	0	0	$\overline{A}BCD$
	01	0	1	1	1	$\overline{A}BCD$
	11	0	1	1	1	$ABCD$
	10	1	0	1	1	$ABCD$
		$\overline{A}BCD$	$\overline{A}BCD$	$\overline{A}BCD$	$\overline{A}BCD$	

(b) Standard SOP:

$$\overline{A}BCD + \overline{A}BCD + \overline{A}BCD + \overline{A}BCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD + ABCD$$

"Don't Care" Conditions

Sometimes a situation arises in which some input variable combinations are not allowed. For example, recall that in the BCD code there are six invalid combinations: 1010, 1011, 1100, 1101, 1110, and 1111. Since these unallowed states will never occur in an application involving the BCD code, they can be treated as "don't care" terms with respect to their effect on the output. That is, for these "don't care" terms either a 1 or a 0 may be assigned to the output: it really does not matter since they will never occur.

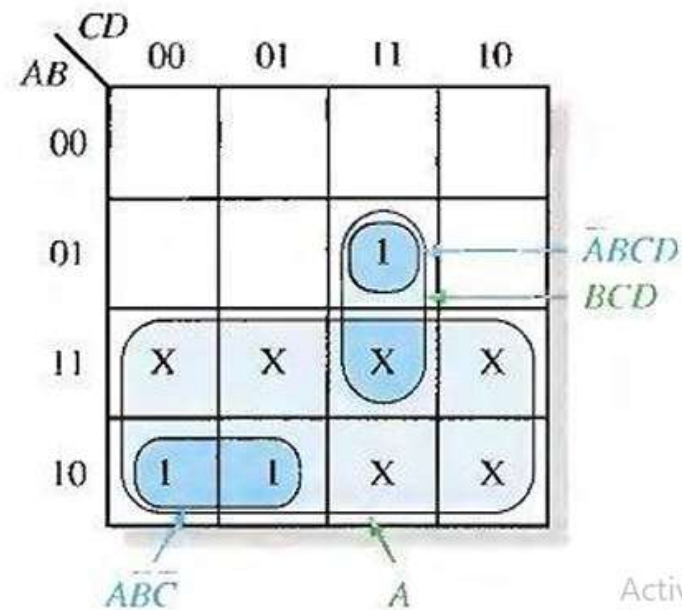
The "don't care" terms can be used to advantage on the Karnaugh map. Fig.(5-9) shows that for each "don't care" term, an X is placed in the cell. When grouping the 1 s, the Xs can be treated as 1s to make a larger grouping or as 0s if they cannot be used to advantage. The larger a group, the simpler the resulting term will be.

The truth table in Fig.(5-9)(a) describes a logic function that has a 1 output only when the BCD code for 7,8, or 9 is present on the inputs. If the "don't cares" are used as 1s, the resulting expression for the function is $A + BCD$, as indicated in part (b). If the "don't cares" are not used as 1s, the resulting expression is $ABC + ABCD$: so you can see the advantage of using "don't care" terms to get the simplest expression.

<i>ABCD</i>	<i>Y</i>
0 0 0 0	0
0 0 0 1	0
0 0 1 0	0
0 0 1 1	0
0 1 0 0	0
0 1 0 1	0
0 1 1 0	0
0 1 1 1	1
1 0 0 0	1
1 0 0 1	1
1 0 1 0	X
1 0 1 1	X
1 1 0 0	X
1 1 0 1	X
1 1 1 0	X
1 1 1 1	X

(a) Truth table

Don't cares



(b) Without "don't cares" $Y = \overline{A}BCD + A\overline{B}\overline{C}$

Activate 'Go to Settings'